SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	26/10/2026	**Duration:**	60 minutes

Code of Conduct

I Divya Dharshini V S (192511308) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

AI-Based Network Intrusion Detection and Threat Monitoring System

1. Analyze System Requirements

1.1 Study the given problem statement and identify the system objectives.

Many organisations will have a computer network where they will hold their data, however there will always be individuals who will try to gain unauthorised access to the network and steal valuable data.

The objectives for this system are:

- Detect any anomalies within the network by monitoring traffic.
- Utilise AI to detect any possible attacks.
- Alert the user if any activities have been found to be suspicious.

- Record any previous alerts and results.
- Allow admins to see any issues with the security and take the appropriate actions.
- Ensure any information within the network is secure.
- Makes monitoring the network traffic easier for admins to see if there are any possible threats.

1.2 Analyze the functional and non-functional requirements that influence the software architecture.

a) Functional Requirements

- Monitor Traffic – Monitor and record network traffic.
- Detect Attacks – Use AI model to analyse data to detect any possible attacks.
- Alert – Alert user if any suspicious activity has been found.
- Dashboard – Display relevant information relating to the network, alerts and attack results.
- Store – Store any alerts and previous results.
- Reports – Allow admins to see/create reports.
- Login – Authorised users can view the system.

These requirements will help define how the components within the system will interact with each other, what data will be shared and how results will be displayed.

b) Non Functional Requirements

- Performance – Quickly analyse network data and display results.
- Security – Ensure network data is secure.
- Reliable – Return useful results and ensure that the system functions correctly.
- Usable – Dashboard is easy to understand.
- Maintainable – Code is easy to update and fix.
- Scalable – System can cope with more network data.
- Availability – System should be available at all times.

These requirements will help define what tools, database, AI model and how the software should be designed.

1.3 Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

The key quality attributes for my project are:

- Scalable – The system can cope with more network traffic.
- Maintainable – Easy for developers to fix any errors and add new features.
- Reliable – The system functions correctly.
- Availability – Admin can use the system.
- Performance – Network data can be analysed and results displayed in a reasonable time frame.
- Security – Only authorised users can view sensitive information.

These quality attributes are important as the system needs to detect any possible attacks and run safely, smoothly and reliably.

2. Select a Suitable Software Architecture

2.1 Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).

For my project, I have decided to use a Layered Architecture. This architectural style means that my system will consist of different layers that will each perform their own tasks to detect potential network attacks and provide the results.

My system will consist of 4 layers:

a) Presentation Layer

This will be what the user can see. This will include the dashboard, network information, alerts and detection results. Using Streamlit I can create a basic and easy to use interface.

b) Application Layer

This layer will handle the main operations of my system. It will receive the data and manage the detection, along with linking everything together.

c) AI Detection Layer

This layer will use the AI model to analyse the network traffic and detect potential attacks. Scikit-learn can be used to create the machine learning model.

d) Data Layer

This layer will store all of the necessary data including network logs, alerts and past detection results. SQLite can be used to store this data. Python and Pandas can also be used to process the network data.

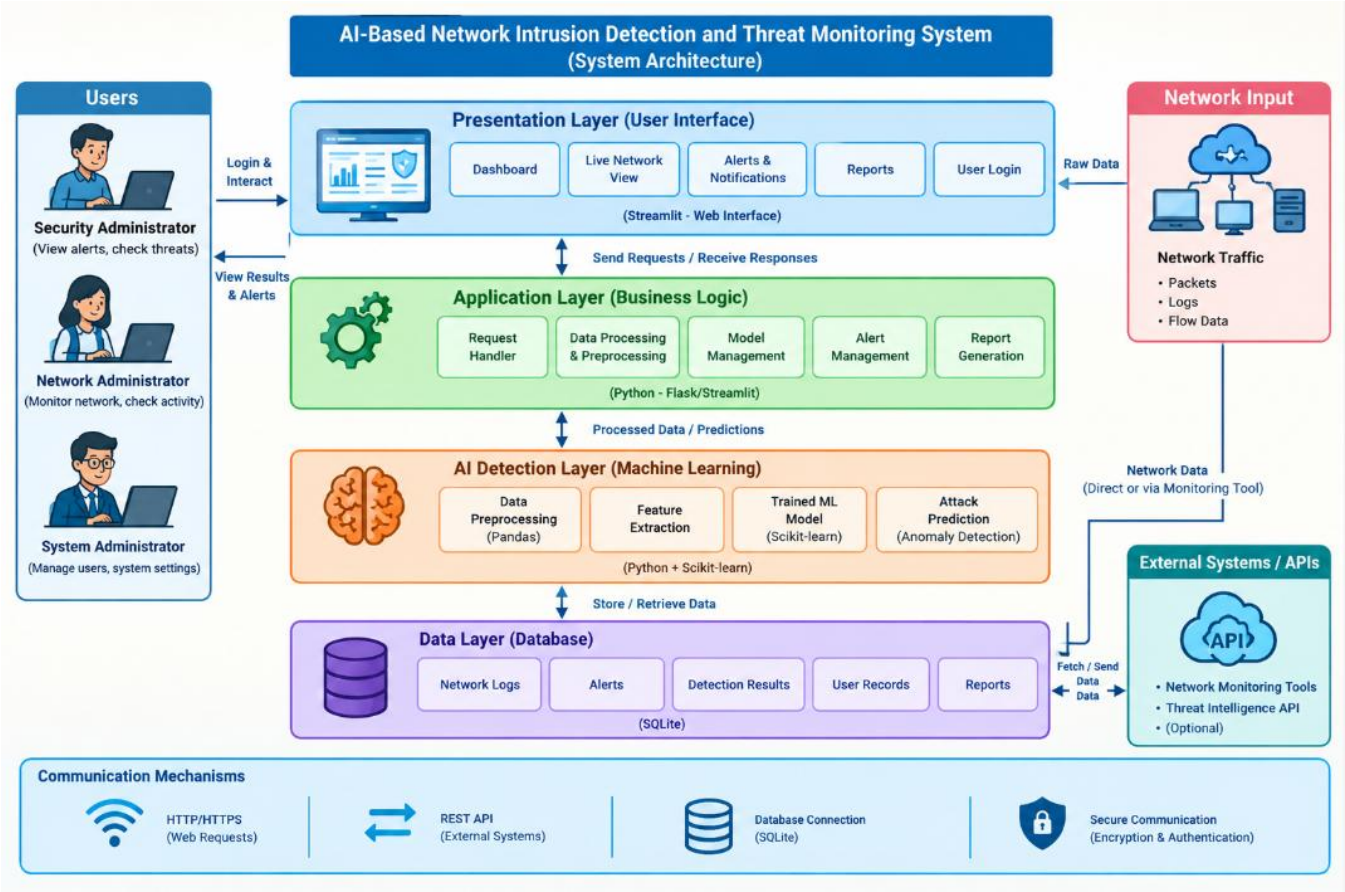
2.2 Justify why the selected architecture is the most suitable for the given problem.

a) Easy to Understand: As there are layers with individual purposes this will make the system easier to understand as the code will be separated.
b) Easy to Maintain: If I need to make any changes to my dashboard or improve the AI model I can just make changes to the specific layer rather than having to completely redesign the system.
c) Better Security: The system can be controlled to ensure the data is transferred between the correct layers. I can also include security checks to keep the network logs secure.
d) Easy to Test: Each layer can be tested individually. For example I can test if the AI model is detecting any suspicious activity before connecting it to the dashbaord.
e) Better Performance Management: The AI detection layer can process the network data, whilst the presentation layer can show the results. This makes it easier to manage the tasks and identify any performance issues.
f) Easy Data Management: The data layer will store any past alerts and results so administrators can look at them when needed.
g) Allows Room for Changes: If I want to add new ways to detect or improve the current features I can easily change the specific layer.
h) Right Choice for My Project: As my project uses python, streamlit, scikit learn, pandas, and SQLite I can use the layered architecture to ensure they all work together in an organised way.

Overall I have decided to use a layered architecture as it will allow my system to be organised and will make it easier to develop, test and improve. It is also a good choice for my project as a student as it doesn't add extra complexity by using a microservice based system.

3. Draw the Software Architecture Diagram

3.1 Design a clear architecture diagram illustrating the overall structure of the proposed system.



3.2 Include all major layers/components, databases, external systems, APIs, and communication mechanisms.

The project will consist of several layers to identify potential network attacks.

a) Presentation Layer : This layer will contain the dashboards, alerts, reports, real-time network and login features. Streamlit will be used to develop the UI.

b) Application Layer : This layer will contain the main functionality of the application including processing requests, preparing data, managing alerts and creating reports. This layer will be developed using Python.

c) AI Detection Layer : This layer will contain the trained Machine Learning model to analyse the network data. Using Pandas to prepare the data and Scikit-learn to perform machine learning.

d) Data Layer : This will contain all the data from the network logs, alerts, detections, user data and reports. SQLite will be used to store and retrieve data.
e) External Systems : There are network monitoring tools which can provide the network traffic data. A threat intelligence API can be used to provide more threat information.
f) Communication : Communication can be done via HTTP/HTTPS through the web interface. Communication with external systems can be done via APIs.

3.3 Use appropriate software architecture notations and labels.

The architecture diagram has been designed to be simple with clear shapes, labels and arrows.

a) Large rectangles : Large rectangles have been used to show the main layers. These include the presentation layer, application layer, AI detection layer and data layer.
b) Small boxes : Small boxes have been used to represent individual components. These include the dashboard, alerts, attack prediction and report.
c) Arrows : Arrows have been included to show the data flow between the users, layers, external systems and database.
d) Cylinder : The cylinder has been included to show the SQLite database that stores the network logs, alerts and detections.
e) User icons : Icons have been used to show the security administrator, network administrator and system administrator.
f) External system icons : Icons have been used to show external systems. Using a cloud and network icon.

4. Explain Components and Their Interactions

4.1 Describe the purpose and responsibilities of each architectural component.

There are various components within my project (AI-Based Network Intrusion Detection System). Each component has a specific responsibility, which overall helps to detect potential network attacks.

a) Presentation Layer

This section is what the user sees. Using streamlit a simple interface has been created to show the dashboard, network details, alerts and detection results.

b) Application Layer

This section handles the main functions of the system. The application will take in data, format requests, manage alerts and help to connect all the components. This was done using Python.

c) AI Detection Layer

This section will analyse the network data. With the help of pandas the data can be formatted. Scikit-learn was used to create and implement the machine learning model. The model will classify if the network activity is normal or suspicious based on its training.

d) Data Layer

This section will store all the important data, this includes network logs, alerts and past detection results. The data is stored using SQLite.

e) External Systems

A network monitoring system can be used to provide data on the network traffic. An external threat intelligence API could be added to provide information on known threats.

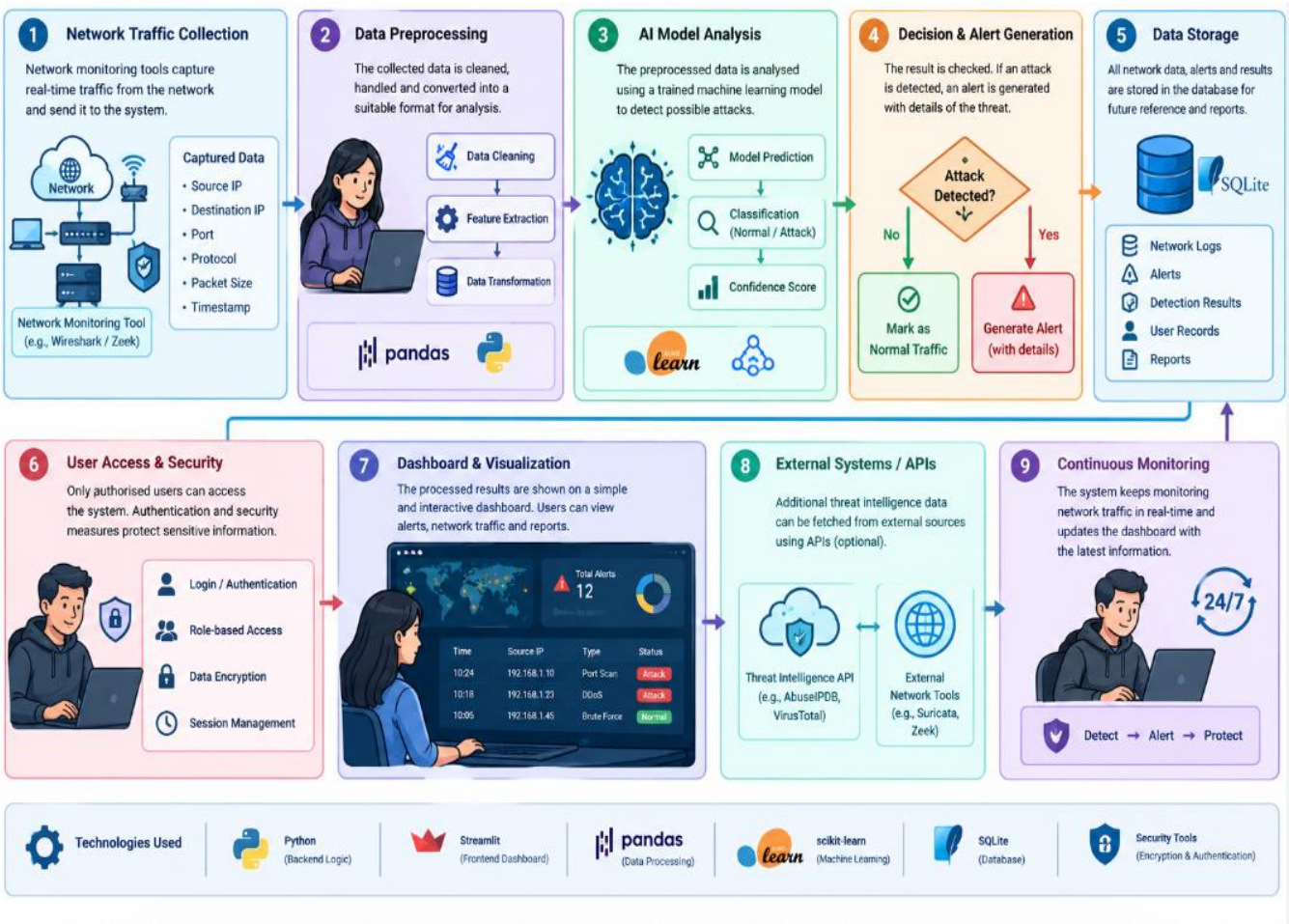
4.2 Explain how the components communicate and exchange data.

Each component will communicate with each other to detect potential network attacks.

- Data on the network traffic is collected from a data source that is available, this will then be sent to the application layer.
- The data will be formatted and sent to the AI detection layer. The AI model will analyse the data and determine if the activity looks normal or suspicious.
- This detection result will then be sent back to the application layer. If suspicious activity is detected an alert will be created and sent to the presentation layer. The administrator can view the alert and details on the dashboard.
- Important data will be stored in the SQLite database. If the administrator wants to view past data, the application will pull through the required information from the database and display it on the dashboard.

- Python functions are used to connect the internal components of the system. Communication can be done using HTTP/HTTPS to send data via web. APIs can be used to send data to external services if needed. Connections to databases are used to store and retrieve data.

4.3 Illustrate the overall workflow of the system.



5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

5.1 Scalability

This shows that additional data can be added into the system. For this project, this can be applied to the AI Detection Layer and Data Layer. This can have more data added which can improve the system. Future improvements can be made by having better hardware or increased processing.

5.2 Security

Security is an important aspect when designing software. In this project there is a login page to ensure only real users can access the dashboard. This can be applied to keep data such as network records secure. HTTPS can be applied to ensure data is secure when being passed. This will help keep data secure within the network.

5.3 Performance

This shows how fast the data can be processed. The network data can be processed using Pandas. After processing, the data can be analysed using the trained AI model with scikit-learn. The Application Layer ensures everything is connected and shown on the dashboard. With the correct data and efficient model suspicious activity can be detected quickly.

5.4 Reliability

This ensures that the system works correctly. The project uses multiple layers to process data, detect attacks, show the results and store records. This makes it easy to test parts of the system. The system should also be able to cope with incorrect data and store important alerts. Constant testing can be carried out to improve reliability.

5.5 Maintainability

This means that changes can be easily made, errors can be fixed and new features added. The system has been designed with layers meaning each layer has a specific purpose. Improvements can be made to the dashboard without affecting the whole AI Detection Layer. If better training data can be found then the machine learning model can be improved. The code is well organised making it easy to maintain.

5.6 Availability

This means the system will be available. The administrators can view the network data and alerts on the dashboard. Ensuring the application is working correctly will help identify any errors. Correct software can be maintained to improve availability. Constant back-ups can be used to protect records that have been stored. This will depend on the computer, network and hosting.

6. Justify Architectural Decisions

Explain the rationale behind your architectural choices.

The architecture has four sections:

- Presentation Layer – Dashboard, alerts and reports.
- Application Layer – Controls core functionality.
- AI Detection Layer – Detects any possible attacks by analysing data from the network.
- Data Layer – Stores any network logs, alerts and detection using SQLite.

Using this architecture will help keep my project well structured and easy to understand. My dashboard can be created using Streamlit, the core functionality will be completed using Python, Pandas will be used to analyse data, Scikit-learn will be used for the Machine Learning and SQLite will be used to store the records.

Discuss the trade-offs considered while selecting the architecture.

There are positives and negatives with each architecture. I have decided to go for a Layered Architecture.

Positives:

- Easy to Understand – Each layer has an obvious role.
- Easy to Test – Each component can be tested.
- Easy to Maintain – Only one layer needs to be changed.
- Easy to develop – It is suited to my project and can be developed using the tools I have available.
- Clear Data Flow – Data flows between the layers clearly.

Negatives:

- May cause delays – There are multiple layers which may cause delays when passing data.
- Does not scale well – If the system needs to handle high volumes of network traffic changes would need to be made to the basic layered application.
- Shared dependencies – If a main component fails it could effect other parts of the application.
- Not easy to scale – It may be difficult to scale individual components.

- Overall the negatives are not too bad and it is easy to develop, is suited to my project and is easy to maintain. If scaling is needed I can improve the design.

Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Layered Architecture will allow me to scale my project.

- Scale – Additional ways of detecting attacks can be added. The AI model can be improved. Further features can be added to the dashboard. In depth reports can be created.
- Integrate – Can integrate with other network monitoring tools or threat intelligence using APIs. This can provide additional information to help identify any potential threats.
- Maintainable – Each layer has a specific role so it will be easy for developers to identify any issues. Individual components can be updated. For example the AI model can be updated without having to update the entire dashboard. The database can also be improved if more storage is needed.
- More Secured – Additional security can be added such as extra logins, user permissions and data protection.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.

Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25